

Supervised Fine-Tuning and Direct Preference Optimization with a Pretrained Base Language Model

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Overview

This project implements supervised fine-tuning (SFT) and direct preference optimization (DPO) on a pretrained causal language model. The objective is to demonstrate correct implementation of prompt/completion formatting, masked causal language modeling loss, sequence log-probability computation, and the DPO objective without relying on high-level training frameworks.

Model and Hardware Setup

- Model: HuggingFaceTB/SmolLM2-360M
- Tokenizer Vocabulary Size: 49,152
- Hardware: Google Colab (Tesla T4 GPU)
- Block Size: 384
- Max Prompt Tokens: 256
- Batch Size: 1

A forward pass confirmed that the model outputs logits of shape $[B, T, V]$, validating correct model loading and configuration.

Dataset

The **Anthropic/hh-rlhf** dataset was used:

- SFT: chosen responses
- DPO: chosen vs rejected pairs

Dataset sizes:

- SFT training examples: 1000
- DPO training examples: 700
- DPO validation examples: 200

Supervised Fine-Tuning (SFT)

Data Processing

Each example was split using the last occurrence of:

`\n\nAssistant:`

- Prompt: everything before and including the marker
- Completion: everything after the marker
- Prompt truncated from the left
- EOS token appended to completion
- Labels:
 - Prompt tokens $\rightarrow -100$
 - Completion tokens $\rightarrow$ supervised
 - Padding $\rightarrow -100$

Loss Function

A manual masked causal language modeling loss was implemented:

- Shifted logits: positions $[0 \text{ to } T - 2]$
- Shifted labels: positions $[1 \text{ to } T - 1]$
- Ignored tokens with -100

Training Result

- Initial loss: ~ 2.25
- Final SFT loss: 1.2344

This indicates successful learning of assistant responses.

Sequence Log-Probability Computation

Sequence log-probabilities were computed manually:

- Applied $\log_{\text{softmax}}$ over logits
- Gathered log-probabilities of target tokens
- Masked non-supervised tokens
- Summed over completion tokens

This was used as the core building block for DPO.

Direct Preference Optimization (DPO)

Dataset Construction

- Extracted prompt/completion for both chosen and rejected responses
- Verified prompt equality
- Tokenized prompt once and reused
- Applied identical truncation
- Masked prompt and padding tokens

Reference Model

A frozen copy of the SFT model was used as the reference model, with gradients disabled.

DPO Objective

The DPO loss was implemented as:

$$-\log \sigma \left(\beta [(\log \pi_{\theta}(y_w|x) - \log \pi_{\theta}(y_l|x)) - (\log \pi_{\text{ref}}(y_w|x) - \log \pi_{\text{ref}}(y_l|x))] \right)$$

with $\beta = 0.1$.

Validation Check (Before Training)

- Initial DPO loss: $0.6914 \approx \log 2$
- Initial reward margin: 0.0

This confirms correct implementation.

Training Results

- Final DPO loss: 0.7070
- Validation loss: 0.6928
- Final reward margin: 0.0125

The small positive reward margin indicates the policy slightly prefers chosen responses over rejected ones.

Qualitative Evaluation

Three prompts were tested on both SFT and DPO models.

Observations

- The SFT model produces coherent but sometimes less structured responses.
- The DPO model shows slight improvement in alignment and helpfulness.
- Due to smaller model size and limited training data, improvements are modest but consistent.

Conclusion

This project successfully demonstrates:

- Correct prompt/completion formatting
- Proper masking strategy for SFT
- Manual implementation of causal LM loss
- Accurate sequence log-probability computation
- Correct DPO objective and training

The results validate that post-training methods such as SFT and DPO can be applied effectively to a pretrained language model, even under constrained computational settings.

References

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